Ideation & Human-Centred Business Model

DS revision sheet: model answers for each checklist item + how to answer from graphs and designs. Based on the course by A. Rekhis.

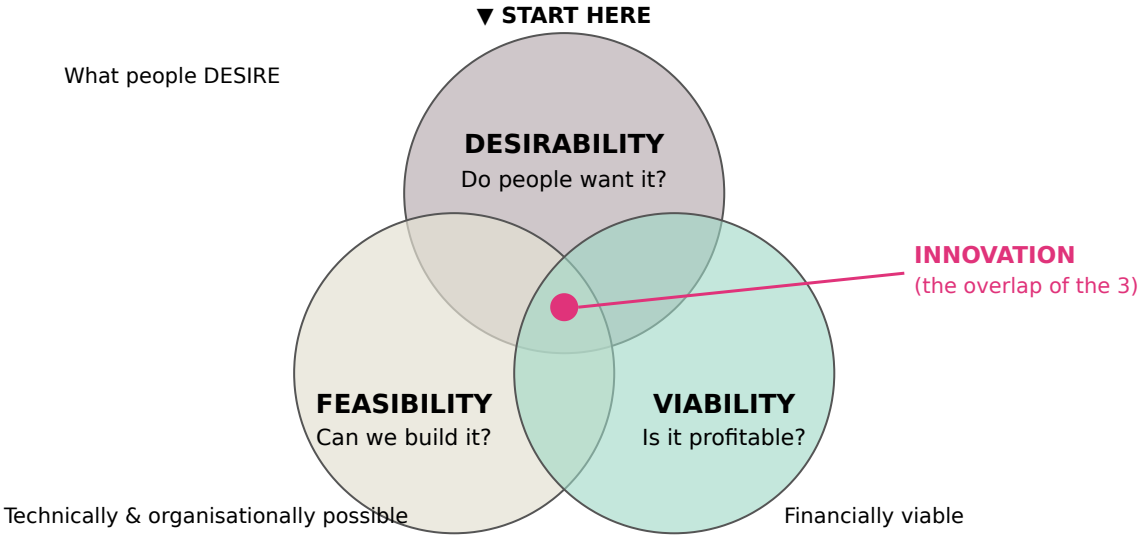
- ☑ **1.** D-F-V, and why we start with Desirability
- ☑ **2.** The 5 phases E-D-I-P-T (iterative / infinity loop)
- ☑ **3.** The 5 W of the Define phase
- ☑ **4.** Brainstorming (7 rules) + Mind mapping
- ☑ **5.** Empathy Map: Says / Thinks / Does / Feels (+ 7-step canvas)
- ☑ **6.** Value Proposition Canvas: Jobs-Pains-Gains ↔ Products-Pain relievers-Gain creators
- ☑ **7.** Flow: Empathy Map → VPC → Business Model Canvas
- ★ **8.** General method: how to answer ANY question about a graph / design + full practical case

How each item below is organised:

- **Model answer**: the text you can write (almost word-for-word) for a theory question.
- **How to draw it**: the diagram to reproduce on your copy, step by step.
- **Trap**: the common mistake to avoid.

Golden rule: in this course, a good answer = **definition** + **diagram** + **explanation of each part** + **a concrete example**.

1 · Desirability – Feasibility – Viability



MODEL ANSWER

An ideal (innovative) solution must sit at the **intersection of three lenses**:

- **Desirability**: what people **desire**. Does the solution meet a real human need? Do users want it?
- **Feasibility**: what is **technically and organisationally possible**. Do we have the technology, skills and resources to build it?
- **Viability**: what is **financially viable**. Can it generate enough revenue to be sustainable?

In the human-centred design process **we always start with Desirability**, because the solution is built around the **human** (the user and their needs). A product that is feasible and profitable but that nobody wants will fail. Once we know what people want, we check whether it can be built (feasibility) and whether it can make money (viability). Solutions at the end of the process must be ***Desirable, Feasible and Viable***.

HOW TO DRAW IT (30 SECONDS)

1. Draw **3 overlapping circles** in a triangle shape: one on top, two below.
2. Top = **Desirability**, bottom-left = **Feasibility**, bottom-right = **Viability**.
3. Write a small arrow **"START HERE"** above Desirability.
4. Mark the **centre (overlap)** with a dot and label it **"Innovation / ideal solution"**.
5. Next to each circle, write its question: *Want it? / Build it? / Profitable?*

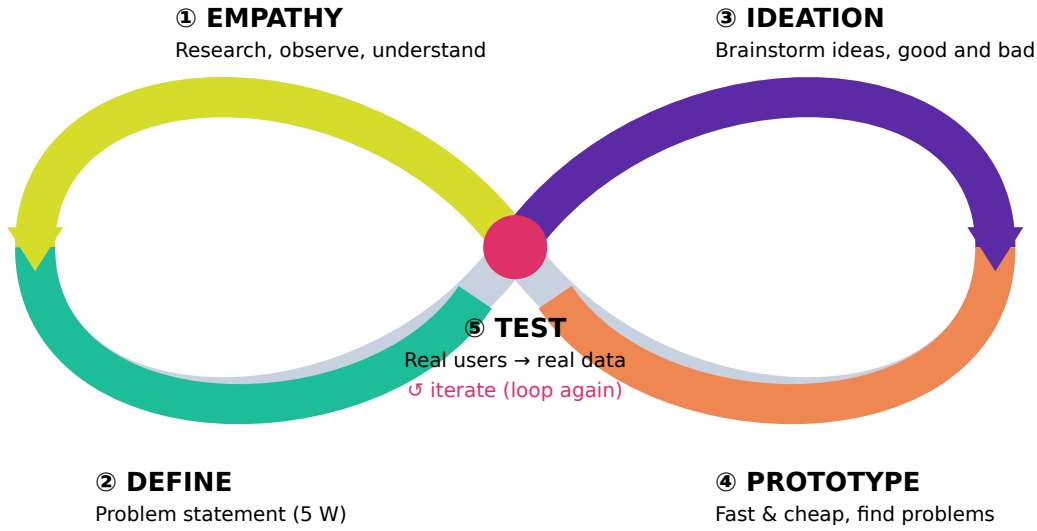
Example (using it in the practical case)

Lens	Example: student housing app
Desirable	Students say they lose time and money finding housing; they want safe, cheap rooms near campus.
Feasible	A simple mobile app + partnership with the university to verify landlords.
Viable	Commission on each rental + paid "premium" listings for landlords.

TRAP

Don't start with "Is it profitable?". The course clearly says **START with Desirability** (the human). And don't confuse Feasibility (technical) with Viability (money).

2 · The 5 phases of Design Thinking: E-D-I-P-T



MODEL ANSWER

Design thinking is a **human-centred** innovation method in 5 phases. It is **iterative**: it is drawn as an **infinity loop (∞)** because after testing we go back to earlier phases and improve.

Phase	What happens	Key words to write
1. Empathy	Designers sit down with **real people** and absorb their points of view, world and introspections **without bias**. Steps: consult **subject-matter experts**, have **in-depth conversations**, **immerse** oneself in the physical environment.	Research, observe, understand, point of view, no bias
2. Define	The designer **integrates the researched understanding** into the human-centric issue and writes a clear **problem statement** (using the 5 W).	Define the challenge, human context, problem statement
3. Ideation	Relies on the **ability to invent**: generate creative solutions to the defined problem. **Think outside the box**; at this stage **don't worry about budget or scalability**. Tools: brainstorming, mind mapping.	Invent, creativity, quantity, outside the box
4. Prototype	Build a first version of the idea: **sketches, models, digital renders**. Fast and cheap, still flexible, used to find problems.	Fast, cheap, flexible
5. Test	**Real users generate real data**. It is not necessarily the last step: because design thinking is built on **iteration**, designers roll out **multiple prototypes**. Without a comprehensive test stage, solutions have **difficulty scaling**.	Real users, real data, iteration, scaling

HOW TO DRAW IT

1. Draw a big ∞ **(infinity sign)**.
2. Left loop: **Empathy** on top → arrow down → **Define** at the bottom.
3. Right loop: **Ideation** on top → arrow down → **Prototype** at the bottom.
4. **Test** in the middle crossing point, with a "↺ iterate" arrow.
5. Write 3–4 words under each phase (see table). If you're short on time, a simple circle with 5 boxes and arrows (E → D → I → P → T → back to E) is also fine, but **always show the loop back**.

TRAP

Don't present it as a straight line that ends at "Test". The key word is **ITERATIVE**. Also remember the order: **E-D-I-P-T** ("**E**very **D**esigner **I**nvents **P**rototypes **T**wice").

Henry Ford quote: *"If I had asked my customers what they wanted, they would have said a faster horse."*
→ **Interpretation:** customers often can't imagine the solution; they only express their need (go faster). This is why **Empathy** means **observing and understanding the deep need**, not just asking "what do you want?". Then **Ideation** invents the real innovation (the car).

3 · The 5 W of the Define phase

MODEL ANSWER

In the **Define** phase, the designer succinctly articulates the problem to solve and writes the **problem statement**. To do it, we answer the **5 W**:

W	Question from the course
WHO	Who are the **stakeholders** affected by the problem?
WHAT	What is the **current state**, the **desired state**, or the **unmet need**?
WHEN	When is the issue occurring, or what is the **timeframe** involved?
WHERE	Where is the problem occurring (a specific **department, location or region**)?
WHY	Why is it **important / worth solving**? How does it impact customers, employees, other stakeholders or the organisation? What is the **magnitude** of the problem? How large is the **gap between the current and desired state**?

HOW TO ANSWER IN THE PRACTICAL CASE

Draw a **5-row table (Who / What / When / Where / Why)**, fill it with the case, then **combine everything into one sentence**. That sentence is your problem statement:

Template: *[WHO] need a way to [WHAT: desired state / unmet need] because [WHY], especially [WHEN] in [WHERE].*

Example: *First-year students (WHO) need affordable and trustworthy housing near campus (WHAT) at the start of each academic year (WHEN) in Tunis (WHERE), because long commutes and scams hurt their grades, budget and well-being (WHY).*

Stakeholders (often asked with "Who")

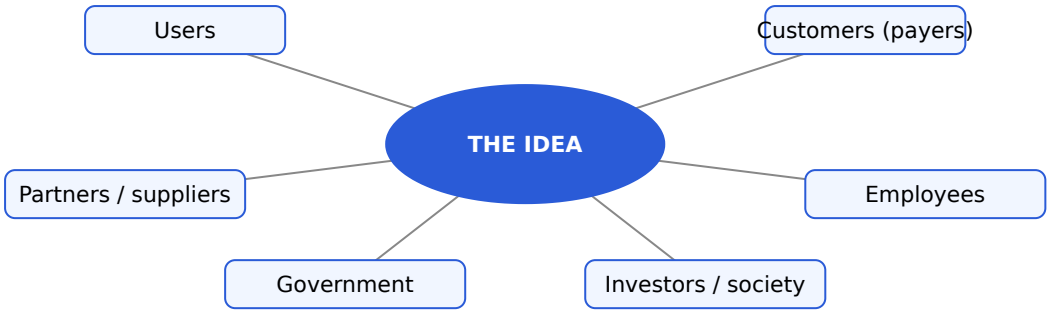
MODEL ANSWER

Stakeholders are **all the people or organisations involved in or affected by the idea**: **users** (who use it), **customers** (who pay; not always the same person as the user), **employees**, **partners and suppliers**, **investors**, **government / regulators**, and **society / the community**.

Prosthesis example from the course: the prosthesis wearer (user), the social security office (payer / bureaucracy), the prosthetist / doctors, suppliers of silicone, family.

HOW TO DRAW STAKEHOLDERS

Put the **idea in a central circle** and draw **arrows/rays** to each stakeholder around it (like a sun). Put the most important ones (users, customers) closest to the centre.


Users
Customers (payers)
THE IDEA
Partners / suppliers
Employees
Government
Investors / society

4 · Brainstorming (7 rules) + Mind Mapping

MODEL ANSWER: BRAINSTORMING

Definition: brainstorming consists of bringing together **a group of people** who will each **share their ideas on a subject**, **without receiving any criticism or judgment** from the others. It is a tool of the **Ideation** phase.

The 7 rules:

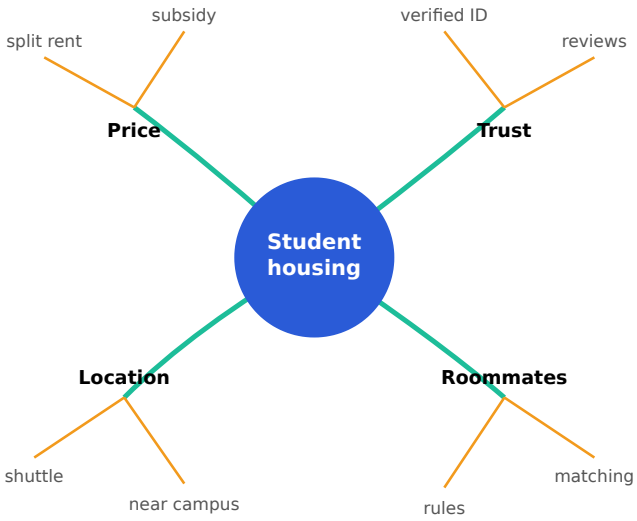
1. **Set a time limit**: keeps energy high and focused.
2. **Begin with a target problem / brief**: everybody works on the same defined problem (from the Define phase).
3. **Refrain from judgment / criticism**: no idea is "bad"; criticism blocks creativity.
4. **Encourage weird and wacky ideas**: crazy ideas can lead to real innovation.
5. **Aim for quantity**: the more ideas, the more chances to find a good one.
6. **Build on others' ideas**: "Yes, and..." rather than "No, but...".
7. **Allow one conversation at a time**: everybody listens and every idea is heard.

Memory trick: "T-T-J-W-Q-B-O": **T**ime, **T**arget, no **J**udgment, **W**acky, **Q**uantity, **B**uild, **O**ne conversation.

MODEL ANSWER: MIND MAPPING

Mind mapping is a **visual representation technique** to **stimulate creativity**. It consists of building a kind of **tree** in which:

- the **main subject is in the centre**;
- **ideas** related to the topic are **branches and sub-branches connected** to it.



HOW TO DRAW IT

1. Write the **problem/subject in a circle** in the middle.
2. Draw 4–5 **main branches** (big themes).
3. Add 2–3 **sub-branches** on each (concrete ideas).

TRAP

Brainstorming ≠ discussion/debate. If a question asks "what is <u>forbidden</u> in brainstorming?" → **criticism and judgment**. If it asks "quality or quantity?" → **quantity**.

5 · Empathy Map: Says / Thinks / Does / Feels

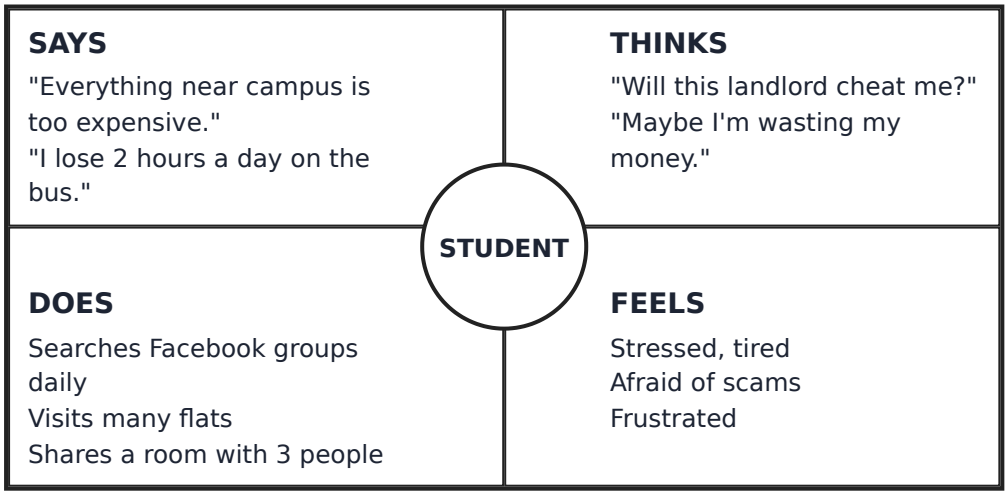
MODEL ANSWER

Definition: empathy maps are **visualization tools** that allow you to **articulate what you know about specific types of users**. They are part of the **design thinking** methodology (Empathy phase) and **create a shared understanding of user needs**. They are usually filled in from **customer interviews** and **observation**.

Quadrant	What it records	Course example
SAYS	What the user **says during an interview**. Capture **exact quotes**.	"I use this product every day because it helps me streamline my workflow."
THINKS	What the user is **thinking throughout the experience**: what occupies their thoughts, what matters to them, their challenges. Key: uncover what they are **too shy or reluctant to say**.	"This feature is really irritating."
DOES	The **actions** the user takes (what you **observe / watch**).	"Keeps refreshing the page."
FEELS	The user's **emotions**: what **worries** them, what **excites** them.	"Excited about the price point" / "Worried it's too hard to use."

HOW TO DRAW IT

Draw a **square divided into 4**, with a **circle "USER" in the centre**. Top-left **SAYS**, top-right **THINKS**, bottom-left **DOES**, bottom-right **FEELS**. Fill each quadrant with 2–4 short "sticky notes" (bullet points).



Example: student looking for housing (same structure as "Jamie buying a TV" in the course)

TRAP: THE 2 CLASSIC CONFUSIONS

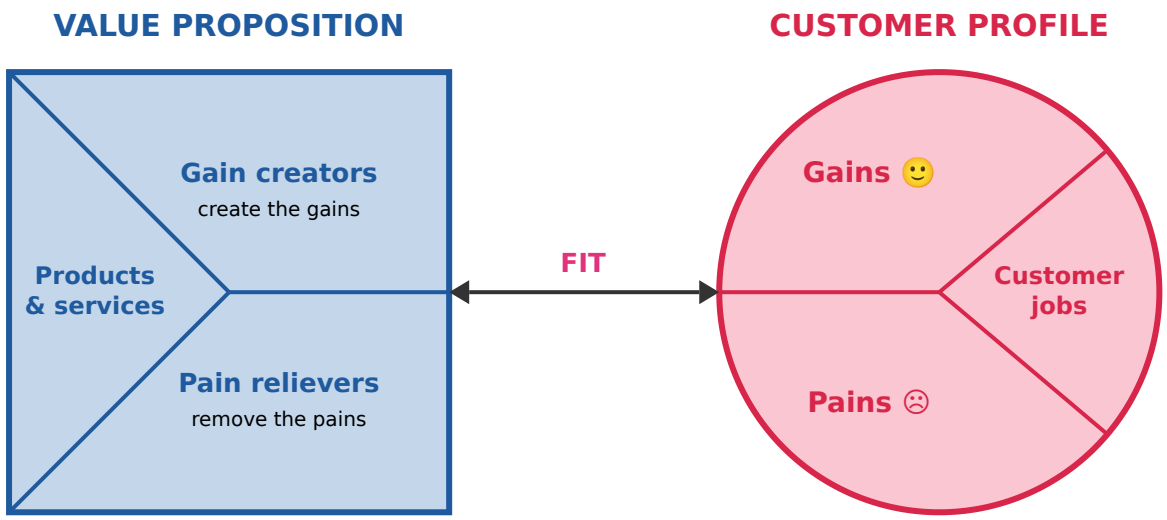
SAYS vs THINKS: Says = said **out loud** (quote in "..."); Thinks = **hidden**, not said.
DOES vs FEELS: Does = an **action** you can see (verb); Feels = an **emotion** (adjective: stressed, excited, afraid...).

The detailed version: Empathy Map Canvas (Dave Gray, 7 steps)

① **WHO** are we empathizing with? Person, situation, role	**GOAL**	② **What do they need to DO?** Job to get done, decision, how they know they succeeded
⑥ **HEAR** From friends, colleagues, second-hand	⑦ **THINK & FEEL** **PAINS** | **GAINS** fears, frustrations, anxieties | wants, needs, hopes, dreams	③ **SEE** Market, environment, what others do ④ **SAY** Quotes heard / imagined
	⑤ **DO** Behaviour today, observed	

Course example (prosthesis wearer in Tunisia): Who: prosthesis wearer in Tunisia · **Need to do:** reduce the frequency of silicone changes · **See:** bionic prostheses, bureaucracy, expensive alternatives · **Say:** "I have to wait more than one year for social security", "It hurts after one year of use" · **Hear:** "You have to wait", "There are alternative solutions without silicone" · **Do:** avoids going out when it's worn out, minimizes prosthesis use · **Pains:** powerlessness, dependency, frustration, weakness, uncertainty, being judged · **Gains:** independence, painless prosthesis, less delay, comfort, freedom.

6 · Value Proposition Canvas (VPC)



MODEL ANSWER

The Value Proposition Canvas is a tool that makes sure the **offer matches the customer's real needs**. It has **two parts**:

1. Customer Profile (circle, right). Always fill it FIRST:

- **Customer jobs**: what the customer is trying to **get done** (tasks, problems to solve, needs).
- **Pains**: negative things: frustrations, fears, risks, obstacles linked to the jobs.
- **Gains**: positive outcomes: what they want, hope for, benefits that would make them happy.

2. Value Proposition / Value Map (square, left):

- **Products & services**: what we offer.
- **Pain relievers**: how our products **eliminate or reduce the pains**.
- **Gain creators**: how our products **produce the gains**.

FIT is achieved when **each pain has a pain reliever** and **each important gain has a gain creator**. The Pains and Gains come directly from the **Empathy Map** (Think & Feel).

HOW TO DRAW IT

1. **Right: a circle** cut in 3 parts: **Gains** (top), **Pains** (bottom), **Customer jobs** (right).
2. **Left: a square** cut in 3 parts: **Gain creators** (top), **Pain relievers** (bottom), **Products & services** (left).
3. Draw a **double arrow between them** and write **"FIT"**.
4. Tip: the parts face each other: **Gains ↔ Gain creators** (top), **Pains ↔ Pain relievers** (bottom), **Jobs ↔ Products** (middle).

Course example (prosthesis) and practical-case example (student housing)

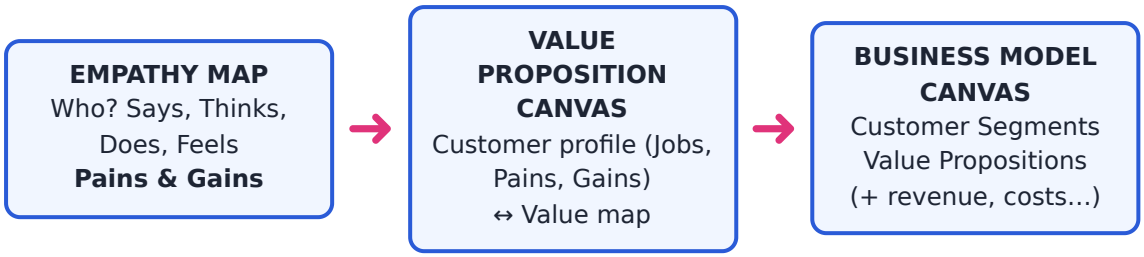
Box	Prosthesis wearer (course)	Student housing (practice)
Customer jobs	Avoid going out when it's worn out; minimize prosthesis use	Find a room, pay rent, get to class
Pains	Powerlessness, dependency, frustration, weakness, uncertainty, being judged	High prices, scams, long commute, stress

Gains	Independence, painless prosthesis, less delay, comfort, freedom	Cheap, close, safe, good roommates
Products & services	New prosthesis solution / service	App with verified listings + roommate matching
Pain relievers	Equality with other people, more autonomy, backup, clear process and delays	Landlords verified by the university, rent-splitting, distance filter
Gain creators	More autonomy, alternative solution (material, backup, temporary solution), backup solution	Student reviews, compatibility matching, secure payment

TRAP

Don't put the company's features in the circle. The circle = **the customer only** (their jobs, pains, gains). The square = **your solution**. And **Pain relievers answer Pains**, **Gain creators answer Gains**; don't mix them.

7 · The flow: Empathy Map → VPC → Business Model Canvas



MODEL ANSWER

The three tools are used **one after another** to build a **human-centred business model**:

1. **Empathy Map**: from **customer interviews** we understand the user (says, thinks, does, feels). Its **"Think & Feel" part gives the Pains and Gains**.
2. **Value Proposition Canvas**: these **Pains and Gains are copied into the Customer Profile** (with the customer jobs, taken from "What do they need to DO?" and "What do they DO?"). Then we design the offer: products, pain relievers, gain creators.
3. **Business Model Canvas**: the **"WHO" of the empathy map becomes the Customer Segments** block, and the **value map of the VPC becomes the Value Propositions** block. Then we complete the rest of the business model (channels, revenue, costs...).

Why this order? Because the business model is **built from the human** (Desirability first), not from the product. It links design thinking (Empathy) to the business (Viability).

Mapping: where each piece of information goes

From the Empathy Map...	...goes into
① WHO are we empathizing with?	BMC → **Customer Segments**
⑦ Think & Feel → **Pains**	VPC → **Pains** (→ answered by Pain relievers)
⑦ Think & Feel → **Gains**	VPC → **Gains** (→ answered by Gain creators)
② Need to DO / ⑤ DO	VPC → **Customer jobs**
VPC value map (products + relievers + creators)	BMC → **Value Propositions**

Business Model Canvas: 9 blocks (just in case): Key Partners · Key Activities · Key Resources · **Value Propositions** · Customer Relationships · Channels · **Customer Segments** · Cost Structure · Revenue Streams. (The course only details the 2 in bold.)

HOW TO DRAW IT

Draw **3 boxes from left to right** (Empathy Map → VPC → BMC) with arrows. On the arrows write what is transferred: **"Pains & Gains"** (first arrow), **"Value proposition + Customer segment"** (second arrow).

8 · How to answer from graphs & designs (any question)

A. When the question SHOWS you a diagram ("Explain this figure")

Use the 5-step method N-D-P-L-E:

1. **N – Name it:** "This figure represents the *Value Proposition Canvas*."
2. **D – Define / purpose:** "It is a tool used to… (one sentence)."
3. **P – Parts:** explain **each part one by one** (each circle, quadrant, box, phase).
4. **L – Links:** explain how the parts connect (arrows, overlap, fit, loop).
5. **E – Example:** give a short concrete example (course example or your own).

If you see…	It is…	Key sentence to write
3 overlapping circles	D-F-V	"An ideal solution is at the overlap: desirable, feasible and viable; we start with desirability."
An infinity loop with 5 icons	Design thinking phases	"5 iterative phases: Empathy, Define, Ideate, Prototype, Test."
Square in 4 with "USER" in the middle	Empathy map	"Visualization tool: Says, Thinks, Does, Feels; it creates a shared understanding of user needs."
Big head with ear, eye, mouth	Empathy map canvas	"7 steps: Who, Need to do, See, Say, Do, Hear, Think & Feel (Pains/Gains)."
Square + circle with an arrow	Value Proposition Canvas	"Value map (products, pain relievers, gain creators) must FIT the customer profile (jobs, pains, gains)."
Big grid with 9 boxes	Business Model Canvas	"Describes how a company creates, delivers and captures value; fed by the VPC."
Centre word with branches	Mind map	"Visual technique to stimulate creativity: subject in the centre, ideas as branches and sub-branches."
Henry Ford "faster horse"	Empathy / innovation	"Customers express needs, not solutions; observe deep needs, then innovate."

B. When the question asks you to BUILD a design (practical use case)

Answer plan (follow this order, each step = one drawing):

1. **Stakeholders**: sun diagram (idea in the centre, actors around).
2. **Empathy Map**: 4 quadrants (or 7-step canvas) filled with the case details. Use quotes for SAYS, verbs for DOES, emotions for FEELS.
3. **Problem statement**: 5 W table + one sentence.
4. **Ideation**: mini mind map or list of 4–5 ideas (respecting brainstorming rules) → **choose one** and justify in 1 line.
5. **Value Proposition Canvas**: copy Pains/Gains from the empathy map, then for **each pain write a reliever** and **each gain a creator**.
6. **Prototype & Test** (2 lines): "Prototype: app mock-up / sketch. Test: with 10 real users, collect data, iterate."
7. **D-F-V check**: one line each: why it's desirable, feasible, viable.

TIPS TO GET MAXIMUM MARKS

- **Use the course words**: "without bias", "real users / real data", "iterative", "shared understanding of user needs", "fit", "start with desirability".
- **Be consistent**: the pains in the VPC must be the **same** as in the empathy map; every reliever must answer a pain.
- **Draw neatly with a ruler** and **label every part**. A labelled diagram + 2 lines of explanation beats a long paragraph.
- Read the case and **underline**: quotes (→ Says), actions (→ Does), emotions (→ Feels), problems (→ Pains), wishes (→ Gains).
- Write **3–5 items per box**; an empty box loses points.

Good luck in your DS! 🍀